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Practical 1: Basic SQL syntax

1. SELECT STATEMENT

   Q1

SALES.RETAIL Settings Open in Workspaces Code Versions

```
SELECT *
FROM RETAIL_SALES_DATASET;
```

Results Chart

	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY
1	1	2023-11-24	CUST001	Male	34	Beauty	3
2	2	2023-02-27	CUST002	Female	26	Clothing	2
3	3	2023-01-13	CUST003	Male	50	Electronics	1
4	4	2023-05-21	CUST004	Male	37	Clothing	1
5	5	2023-05-06	CUST005	Male	30	Beauty	2
6	6	2023-04-25	CUST006	Female	45	Beauty	1
7	7	2023-03-13	CUST007	Male	46	Clothing	2

Q2

```
SELECT TRANSACTION_ID,date,Customer_ID
FROM RETAIL_SALES_DATASET;
```

Results Chart

	TRANSACTION_ID	DATE	CUSTOMER_ID
1	1	2023-11-24	CUST001
2	2	2023-02-27	CUST002
3	3	2023-01-13	CUST003
4	4	2023-05-21	CUST004
5	5	2023-05-06	CUST005
6	6	2023-04-25	CUST006
7	7	2023-03-13	CUST007

2.SELECT DISNTICT Statement

Q3

```sql
SELECT DISTINCT Product_Category
FROM RETAIL_SALES_DATASET;
```

Results | Chart

	PRODUCT_CATEGORY
1	Clothing
2	Beauty
3	Electronics

Q4

```sql
SELECT DISTINCT Gender
FROM RETAIL_SALES_DATASET;
```

Results | Chart

	GENDER
1	Male
2	Female

3.WHERE Clause

Q5

```sql
SELECT *
FROM RETAIL_SALES_DATASET
WHERE AGE > 40;
```

Results | Chart

	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY	PRICE_PER_UNIT	TOTAL_AMOUNT
1	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
2	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
3	7	2023-03-13	CUST007	Male	46	Clothing	2	25	50
4	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
5	10	2023-10-07	CUST010	Female	52	Clothing	4	50	200
6	14	2023-01-17	CUST014	Male	64	Clothing	4	30	120
7	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000

Q6

```sql
SELECT *
FROM RETAIL_SALES_DATASET
WHERE PRICE_PER_UNIT BETWEEN 100 AND 500;
```

Results | Chart

	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY	PRICE_PER_UNIT	TOTAL_AMOUNT
1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
2	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
3	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
4	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
5	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
6	16	2023-02-17	CUST016	Male	19	Clothing	3	500	1500
7	20	2023-11-05	CUST020	Male	22	Clothing	3	300	900

Q7

```sql
SELECT *
FROM RETAIL_SALES_DATASET
WHERE PRODUCT_CATEGORY IN ('Beauty','Electronics');
```

Results | Chart

	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY	PRICE_PER_UNIT	TOTAL_AMOUNT
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100
4	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
5	8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
6	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
7	12	2023-10-30	CUST012	Male	35	Beauty	3	25	75

Q8

```sql
SELECT *
FROM RETAIL_SALES_DATASET
WHERE PRODUCT_CATEGORY <> 'Clothing';
```

Results | Chart

	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY	PRICE_PER_UNIT	TOTAL_AMOUNT
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100
4	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
5	8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
6	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
7	12	2023-10-30	CUST012	Male	35	Beauty	3	25	75

Q9

```sql
SELECT *
FROM RETAIL_SALES_DATASET
WHERE QUANTITY >= 3;
```

Results | Chart

	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY	PRICE_PER_UNIT	TOTAL_AMOUNT
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
3	10	2023-10-07	CUST010	Female	52	Clothing	4	50	200
4	12	2023-10-30	CUST012	Male	35	Beauty	3	25	75
5	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
6	14	2023-01-17	CUST014	Male	64	Clothing	4	30	120
7	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000

4. Aggregate Functions

Q10

```sql
SELECT COUNT (*) AS Total_Transactions
FROM RETAIL_SALES_DATASET;
```

Results | Chart

	TOTAL_TRANSACTIONS
1	1000

Q11

```sql
SELECT AVG (Age) AS Average_Age
FROM RETAIL_SALES_DATASET;
```

	AVERAGE_AGE
1	41.392000

Q12

```sql
SELECT SUM(Quantity) AS Total_Quantity
FROM RETAIL_SALES_DATASET;
```

	TOTAL_QUANTITY
1	2514

Q13

```sql
SELECT MAX (Total_Amount)AS Max_Total_Amount
FROM RETAIL_SALES_DATASET;
```

	MAX_TOTAL_AMOUNT
1	2000

Q14

```sql
SELECT MIN (Price_per_unit) AS Min_price_per_unit
FROM RETAIL_SALES_DATASET;
```

	MIN_PRICE_PER_UNIT
1	25

5.GROUP BY Statement

Q15

```sql
SELECT Product_category,COUNT(*)AS Transaction_count
FROM RETAIL_SALES_DATASET
GROUP BY PRODUCT_CATEGORY;
```

	PRODUCT_CATEGORY	TRANSACTION_COUNT
1	Clothing	351
2	Beauty	307
3	Electronics	342

Q16

```sql
SELECT Gender,SUM (Total_Amount)AS Total_Revenue
FROM RETAIL_SALES_DATASET
GROUP BY GENDER;
```

Results Chart

	GENDER	TOTAL_REVENUE
1	Male	223160
2	Female	232840

Q17

```sql
SELECT Product_category,AVG(Price_per_unit)AS Average_Price
FROM RETAIL_SALES_DATASET
GROUP BY PRODUCT_CATEGORY;
```

Results Chart

	PRODUCT_CATEGORY	AVERAGE_PRICE
1	Beauty	184.055375
2	Clothing	174.287749
3	Electronics	181.900585

6.HAVING CLAUSE

Q18

```sql
SELECT Product_category,SUM(Total_Amount) AS Total_Revenue
FROM RETAIL_SALES_DATASET
GROUP BY PRODUCT_CATEGORY
HAVING SUM (Total_Amount)>10000;
```

Results Chart

	PRODUCT_CATEGORY	TOTAL_REVENUE
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Q19

```sql
SELECT Product_category,AVG(Quantity) AS Average_Quantity
FROM RETAIL_SALES_DATASET
GROUP BY PRODUCT_CATEGORY
HAVING AVG (QUANTITY)>2;
```

Results Chart

	PRODUCT_CATEGORY	AVERAGE_QUANTITY
1	Beauty	2.511401
2	Clothing	2.547009
3	Electronics	2.482456

7.CASING Statement

Q20

```sql
SELECT
Transaction_ID,
Total_Amount,
CASE
WHEN Total_Amount >1000 THEN 'High'
ELSE 'Low'
END AS Spending_Level
FROM RETAIL_SALES_DATASET;
```

Results Chart

	TRANSACTION_ID	TOTAL_AMOUNT	SPENDING_LEVEL
1	1	150	Low
2	2	1000	Low
3	3	30	Low
4	4	500	Low
5	5	100	Low
6	6	30	Low
7	7	50	Low

Q21

```sql
SELECT
Customer_ID,
Age,
CASE
WHEN Age <30 THEN 'Youth'
WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
ELSE 'Senior'
END AS Age_Group
FROM RETAIL_SALES_DATASET;
```

Results Chart

	CUSTOMER_ID	AGE	AGE_GROUP
1	CUST001	34	Adult
2	CUST002	26	Youth
3	CUST003	50	Adult
4	CUST004	37	Adult
5	CUST005	30	Adult
6	CUST006	45	Adult